Applying Probability and Statistics to Evaluate a Machine Learning Model

Logistic Regression vs Random Forest on the Breast Cancer Wisconsin Diagnostic Dataset

1. Problem Statement and Dataset

Problem. Early and accurate classification of breast tumours as malignant or benign is a high-stakes task. The goal here is to decide, using probability and statistics, whether a more flexible model (Random Forest) is genuinely better than a simple baseline (Logistic Regression), or whether any observed difference in test accuracy could just be due to chance in a finite test sample.

Research question. Do Logistic Regression and Random Forest have different probabilities of correctly classifying a tumour from the same population?

Dataset

Property	Value
Source	Breast Cancer Wisconsin Diagnostic (scikit-learn, load_breast_cancer)
Rows (patients/samples)	569
Features	30 numeric features computed from cell-nuclei images (mean, standard error and worst value of 10 measurements)
Target	0 = malignant, 1 = benign
Class counts	Benign: 357 (62.7%); Malignant: 212 (37.3%)

Output: dataset overview (screenshot)

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================================================================================
DATASET OVERVIEW
================================================================================
Dataset: Breast Cancer Wisconsin Diagnostic
Number of rows: 569
Number of features: 30
Target mapping: 0 = malignant, 1 = benign
Class counts:
target
benign       357
malignant    212

Training rows: 426
Held-out test rows: 143
Sampling method: stratified random train/test split (75%/25%)
Random seed: 42
```

2. Sampling and Experimental Methodology

2.1 Sampling

A **stratified random hold-out split** was used: 75% of the data (426 rows) for training and 25% (143 rows) for testing, with random seed 42 for reproducibility. Stratification keeps the benign/malignant proportion approximately equal in both subsets, so the test set is a representative random sample of the population. Both models are evaluated on the **exact same 143 test rows**, which makes the comparison paired.

2.2 Models

- **Logistic Regression (baseline):** StandardScaler followed by LogisticRegression (max_iter = 5000), wrapped in a Pipeline so that scaling is fitted on training data only (no data leakage).
- **Random Forest (comparison):** 300 trees, random_state = 42. No scaling is needed for tree models.

2.3 Evaluation metrics

Accuracy is computed on the test set. Precision, recall (sensitivity) and F1 are computed with **malignant (label 0) as the positive class**, since missing a malignant tumour is the costly error.

2.4 Statistical procedures

- **95% Wilson confidence interval** for each model's accuracy, treating the number of correct predictions as a Binomial(n = 143, p) random variable.
- **Exact two-sided McNemar test** (binomial form) on paired correctness outcomes, with H_0: both models have the same probability of being correct on a test case; H_1: the probabilities differ. Significance level $\alpha = 0.05$.

3. Observations, Decision, and Limitations

3.1 Model results

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Logistic Regression
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Correct predictions: 141/143
Accuracy: 0.9860 (98.60%)
95% Wilson CI for accuracy: [0.9504, 0.9962]
Confusion matrix (rows=true, columns=predicted; labels [malignant, benign]):
[[52  1]
 [ 1 89]]
Malignant precision: 0.9811
Malignant recall/sensitivity: 0.9811
Malignant F1-score: 0.9811

Random Forest
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Correct predictions: 137/143
Accuracy: 0.9580 (95.80%)
95% Wilson CI for accuracy: [0.9115, 0.9806]
Confusion matrix (rows=true, columns=predicted; labels [malignant, benign]):
[[49  4]
 [ 2 88]]
Malignant precision: 0.9608
Malignant recall/sensitivity: 0.9245
Malignant F1-score: 0.9423
```

Comparison table

Model	Correct	Accuracy	95% CI (Wilson)	Malignant precision	Malignant recall	Malignant F1
Logistic Regression	141 / 143	0.9860	[0.9504, 0.9962]	0.9811	0.9811	0.9811
Random Forest	137 / 143	0.9580	[0.9115, 0.9806]	0.9608	0.9245	0.9423

Confusion matrices (rows = true, columns = predicted)

	LR: pred. malignant	LR: pred. benign	RF: pred. malignant	RF: pred. benign
True malignant	52	1	49	4
True benign	1	89	2	88

Logistic Regression made 2 errors (1 false negative, 1 false positive); Random Forest made 6 errors (4 false negatives, 2 false positives). Notably, Random Forest missed 4 of 53 malignant tumours, versus 1 for Logistic Regression.

3.2 Paired hypothesis test (exact McNemar) - screenshot

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PAIRED HYPOTHESIS TEST: EXACT McNEMAR TEST
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H0: The two models have the same probability of being correct on a test case.
H1: The models differ in their probability of being correct on a test case.
Correct only for Logistic Regression: 5
Correct only for Random Forest: 1
Total discordant test cases: 6
Two-sided exact p-value: 0.2188
Significance level alpha: 0.05
Decision: Fail to reject H0. The difference is not statistically significant at 5%.
Interpretation: The observed accuracy difference could plausibly reflect test-sample variation.

Saved results table to: model_comparison_results.csv

Important limitation: this is one fixed hold-out split. Results may change with
another split; use repeated stratified cross-validation for a more robust follow-up.
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Paired outcome table	RF correct	RF wrong
LR correct	136	5 (LR only)
LR wrong	1 (RF only)	1

3.3 Decision

The p-value (0.2188) is greater than $\alpha = 0.05$, so we **fail to reject H_0**. Although Logistic Regression scored higher (98.60% vs 95.80%, a gap of 2.8 percentage points), this evidence is not strong enough to conclude that one model is truly better on the underlying population. The two 95% confidence intervals ([0.9504, 0.9962] and [0.9115, 0.9806]) overlap substantially, which is consistent with this conclusion.

Practical conclusion: both models perform very well. Since the simple, interpretable Logistic Regression is at least as accurate and does not show a statistically significant disadvantage, it is a sensible choice for this dataset. Failing to reject H_0 does *not* prove the models are equal; it means the test set is too small to separate them.

3.4 Limitations

- **Single fixed split.** Results depend on one random split (seed 42) and may change with another.
- **Small test set.** With n = 143 and only 6 discordant cases, the McNemar test has low statistical power; a real but small difference could go undetected.
- **No hyperparameter tuning.** Both models use near-default settings; tuning could shift the ranking.
- **Single dataset.** The findings may not generalise to other populations, hospitals or imaging protocols.
- **Accuracy alone is incomplete.** In a medical setting, recall on malignant cases matters most; costs of false negatives and false positives are not equal.
- **Follow-up.** Repeated stratified k-fold cross-validation with a corrected paired test would give a more robust comparison.

4. Where Probability and Statistics Are Used

Concept	Where it appears	Role in this study
Population and random sampling	Stratified random 75/25 split	The test set is a random sample from the population of tumours, so test performance is an estimate of true performance.
Class probabilities (prior)	Benign 357/569 = 62.7%, malignant 37.3%	Describes class balance; stratification preserves these proportions in train and test sets.
Probabilistic model	Logistic Regression	Models P(malignant | features) via the logistic (sigmoid) function; classification is a probability threshold.
Ensemble voting / randomness	Random Forest (300 trees)	Bootstrap sampling and random feature selection; the final class is the majority vote of the trees.
Binomial distribution	Number of correct predictions out of 143	Each prediction is a Bernoulli trial, so the correct count is Binomial(n, p); accuracy is the estimator of p.
Confidence interval	95% Wilson interval	Quantifies uncertainty in accuracy: LR [0.9504, 0.9962], RF [0.9115, 0.9806].
Hypothesis testing	Exact McNemar test	Tests H0 (equal probability of being correct) using the 6 discordant pairs; p-value = 0.2188.
Conditional probability metrics	Precision, recall, F1	Precision = P(malignant | predicted malignant); recall = P(predicted malignant | malignant).
Statistical significance / Type I error	$\alpha = 0.05$	Limits the probability of wrongly declaring a difference when none exists to 5%.

How the p-value is obtained

Under H_0, each of the 6 discordant cases is equally likely to favour either model, so the number of cases favouring Random Forest is Binomial(6, 0.5). The observed minimum count is 1, giving $p = 2 \times P(X \leq 1) = 2 \times (1 + 6)/64 = 14/64 = 0.2188$.

Reproducibility

Run `pip install numpy pandas scikit-learn scipy` and then `python probability_statistics_model_comparison.py`. The script saves `model_comparison_results.csv`.